VINUNIVERSITY

Test duration: 1 hour and 30 minutes

Name: **Student ID:**

Midterm Practice Test 3 – Fall 2026

COMP3040/5030 Computer Vision Unofficial practice test, prepared by Lê Tăng Phú Quý

Instructions:

- Academic honesty is required in all work you submit to be graded. You should **NOT** copy or share your solutions with other students to avoid plagiarism issues.
- For short answer questions, please make your answers as **concise** as possible.
- This is a closed-book exam and NO Internet access is allowed. You are allowed 1 page of **hand-written** note (both sides).
- Drafting paper and pen/pencil are allowed. We will provide the drafting paper to you.
- Late submission without an approved extension will not be graded at all.
- You must **RETURN** this hard copy of the midterm exam at the end of the exam period.
- During the exam, you are **NOT** allowed to leave the room or/and use the restroom.
- You are required to **NOT** use your mobile phone in the exam room. Please put it on the table or in your bag.
- Good luck!

About the exam: There are **05** problems with **100 points** total, and some have multiple parts. Please check through the pages before you start.

Problem	Content	Point (Undergraduates)	Type
1	Fundamentals of Computer Vision	20	short answer
2	Image Filtering	25	long answer
3	Feature Detection and Matching	25	long answer
4	Fitting and Image Transformations	25	long answer
5	System Design	05	long answer
Total		**100**	

Problem 1. Fundamentals of Computer Vision

Give a short answer for each of the following question:

1.1. A 2 m pole stands 10 m from a camera with a 35 mm focal length. How tall is its image (mm)? At 20 m? Why can a single image not reveal distance? (5 pts)

1.2. `a = np.uint8(200)`. What do `a + np.uint8(100)` and `cv2.add(a, 100)` give? A 2×2 average pooling of 1, 2, 2, 2 stored into a `uint8` array gives what? (5 pts)

1.3. Minimum point correspondences to estimate: a translation, a similarity, an affine map, a homography? State the general rule. (5 pts)

1.4. An 8-bit image has an MSE of 25 against the original. Compute the PSNR. How must the noise change to gain 6 dB? (5 pts)

Problem 2. Image Filtering

2.1. Four pixels $F(0,0) = 10$, $F(1,0) = 20$, $F(0,1) = 30$, $F(1,1) = 50$ (arguments (x, y)). (a) Bilinear interpolation at $(0.5, 0.5)$ and $(0.25, 0.75)$. (b) Upscale $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ ×2 with nearest neighbour. (10 pts)

2.2. Each row of a 7×7 image is $[0, 0, 0, 0, 10, 10, 10]$ (a vertical step edge). Correlate horizontally with $[-1, 0, 1]$. (a) Compute one output row with **zero padding**. (b) What does the value in the last column mean? Redo with **replicate padding**. (c) How many local maxima are in the output of (b)? (15 pts)

Problem 3. Feature Detection and Matching

3.1. A Harris window (uniform weights) contains 4 gradients (I_x, I_y): $(10, 0), (0, 10), (10, 10), (-10, 10)$. (a) Compute M. (b) With $k = 0.05$ compute R and classify. (c) Redo with $(10, 0), (20, 0), (10, 0), (-10, 0)$. (13 pts)

3.2. 1-D template $T = [1, 2, 3]$, image $I = [0, 2, 4, 6, 3, 1]$. (a) Compute the SSD map ('valid') and the best position. (b) Compute the ZNCC at positions 0 and 1. (c) Comment on the ZNCC result. (12 pts)

Problem 4. Fitting and Image Transformations

4.1. The estimated inlier ratio is $w = 0.7$ and we want $p = 0.99$. Compute the minimum number of RANSAC iterations for: a line, an affine map, a homography. Comment. (12 pts)

4.2. A 400×300 image (width × height). As in the Lec07 lab: shift the image 100 px right and 100 px down, then rotate $45°$ counter-clockwise about the image centre and scale ×2 (with `cv2.getRotationMatrix2D`). (a) Write both 2×3 matrices and the composed one. (b) Where does pixel $(0, 0)$ of the original go? (13 pts)

Problem 5. System Design

5. A lab needs to **count and size** nearly circular cells/coins in microscope images (noisy, some touching). Design the algorithm and explain how to choose its parameters. (5 pts)